

1. Verification plan for adder-subtractor: (4-bit)

> objectives:

i) For addition:

- Check the sum
- Check the carry out.

ii) For subtraction:

- Check for difference.
- If $a > b$, check for $a - b$.
- If $a < b$, check for $b - a$.
- Check for borrow.
- Check for $cin = 0$, $cin = 1$.

> Checking strategy:

- Check for corner cases of extremes like 0 & 15.
- Check for intermediate values of transitions like 3, 4, 7, 8.
- Compare the OUT response & the expected result in scoreboard.

> Assertions:

- Check for sum and difference. Do not let them be in X or Z.
- Inputs should not be in X or Z.

> Exit strategy:-

100% code & functional coverage. 0% errors, warnings & failures.